

Data management for LLM training

A clean, evidence-first pipeline for Vietnamese dataset curation.

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WHY LLMS + VIETNAMESE

Vietnam needs LLMs that speak Vietnamese to preserve meaning, culture, and local trust

- **The Power of LLMs:** Vietnamese AI can scale local reasoning, language understanding, and real-time knowledge work.
- **Cultural Precision:** Native models preserve tone, idioms, and cultural meaning without mistranslation.
- **Data Sovereignty:** Keeping AI infrastructure local protects citizen, corporate, and government data.
- **Local impact:** Tailored models accelerate healthcare, education, legal services, and public administration.

DATA NECESSITY

Why Vietnam lags behind the US in LLM training data

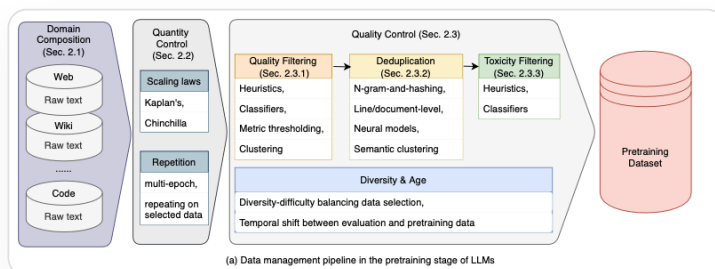
- **Low-resource gap:** Vietnam has far less digitized, labeled, and open Vietnamese text than the US.
- **Hidden content:** Valuable local sources are often private, government-owned, or not prepared for AI ingestion.
- **Infrastructure gap:** Vietnam needs local pipelines, curation standards, and shared datasets to build native models.

What must change: collect and curate Vietnamese corpora, standardize local data formats, and build provenance-rich pipelines so native models can learn from real language, culture, and public-domain knowledge.

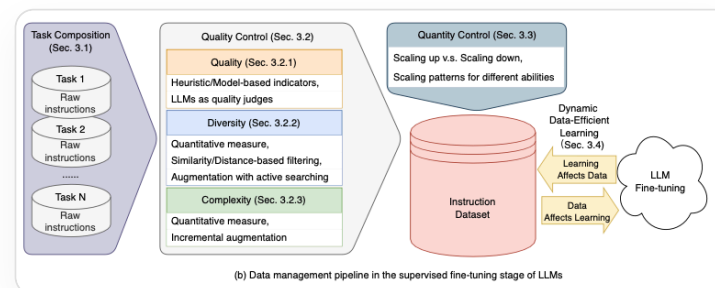
TRAINING STAGES

What data is needed in LLM training?

- **Pretraining:** Broad Vietnamese corpora and multilingual context to teach language patterns.
- **Fine-tuning:** Curated Vietnamese examples, QA pairs, domain documents, and safety-aligned samples.



Pretraining data pipeline

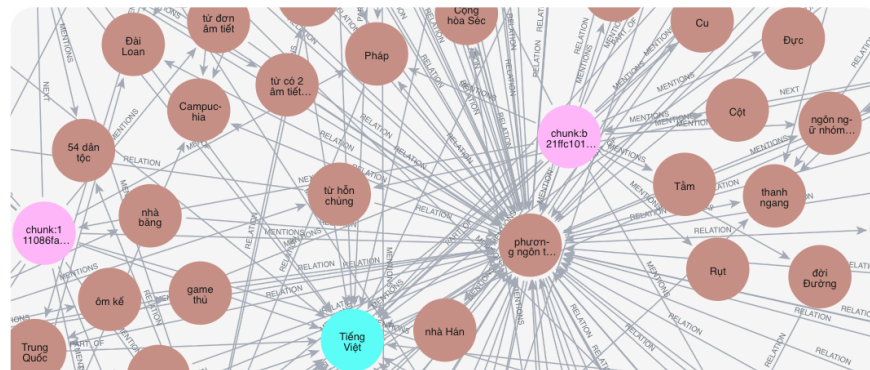


Fine-tuning data pipeline

All data must be cleaned, deduplicated, and filtered to remove bad, noisy, or irrelevant content before it enters the pipeline.

OUR SOLUTION

Why knowledge graphs make Vietnamese LLM training more efficient

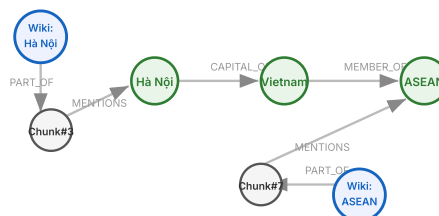


- Knowledge graphs capture entities, relationships, and provenance in a structured form.
- That structure gives LLMs clearer, higher-value training signals than raw text alone.
- Graphs make it easier to generate grounded fine-tuning examples and support retrieval-based reasoning.
- They also help reuse knowledge across tasks while preserving accuracy and explainability.

KG → SFT DATA

Knowledge graphs unlock reasoning that spans documents

- Entities connected by relationships let us **trace paths across documents** — e.g. "Which university did the author of paper X attend?" requires hopping from paper → author → education.
- Without a graph, generating multi-hop QA pairs is **prohibitively hard** — you'd need to manually cross-reference every document pair.



Q: Which regional organization does Vietnam belong to?

A: Vietnam is a member of ASEAN — sourced from *Wiki:ASEAN* (chunk #7) and linked via *Wiki:Hà Nội* (chunk #3).

The graph joins two independent Wikipedia articles through a shared entity (ASEAN), enabling a QA pair that no single document could answer.

KG CHARACTERISTICS

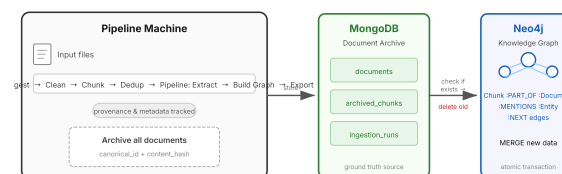
What We Focused On

- **Scalability:** LLMs require vast amounts of data to perform well, and new data is constantly being generated. We therefore store the graph in a graph database (Neo4j) to support growing number of nodes and relations. And we implement a legal web scraper to continuously feed new data since the LLMs can never get enough data.
- **High-quality data:** LLMs need high quality data to perform well, and to ensure the data is accurate and reliable with no hallucinations. We evaluate with metrics (completeness, consistency, duplication rate, factual grounding) and compare fine-tuning LLMs with KG data vs without KG vs no fine-tuning.
- **Support for evolving knowledge:** We want to be able to update the knowledge graph as new information becomes available. When this happens, we might get a new version of an existing document, we then delete that document from the knowledge graph, but store it in a separate Database. keep historical versions of source documents while evolving the KG by iteratively adding nodes and evidence.

These characteristics ensure the KG is robust, auditable, and ready to drive both model training and runtime retrieval strategies.

PIPELINE ARCHITECTURE

How our Vietnamese KG pipeline flows



The knowledge graph is exported to, **GraphML**, and **Neo4j CSV** for downstream training pipelines. For large-scale graphs that would exhaust pipeline machine RAM, we stream entities and triples directly into Neo4j — building the graph in-database rather than in memory. Old chunks are atomically replaced when documents are updated.

PIPELINE — DATA PREPARATION

Ingest, clean, chunk, and deduplicate

1. **Ingest:** Load Vietnamese text from web scrapes, Wikipedia dumps, and curated JSONL sources — normalize into a unified document model with metadata.
2. **Clean:** Strip HTML, normalize Unicode (Vietnamese tone marks), remove boilerplate, and filter non-Vietnamese or low-quality content.
3. **Chunk:** Segment documents into semantic chunks (sentence, paragraph, or fixed-size) with stable chunk IDs for traceability.
4. **Deduplicate & QA:** Near-deduplicate at document and chunk level; apply heuristic quality filters to remove noise before extraction.

PIPELINE — KNOWLEDGE EXTRACTION

Extract, resolve, build, and expand the graph

1. **Extract entities & relations:** Identify people, places, organizations, and facts from cleaned Vietnamese text using ML extractors (with optional LLM fallback).
2. **Entity resolution:** Merge mentions of the same real-world entity across documents using embedding similarity and rule-based linking.
3. **Build & validate graph:** Assemble entities and triples into Neo4j with ontology-constrained node types (:Chunk, :Document, :Entity) and relationships (:PART_OF, :MENTIONS, :NEXT).
4. **Export & expand:** Produce JSON, GraphML, Neo4j CSV, and RDF exports; continuously ingest new sources and refresh the graph with full provenance tracking.

EVALUATION — KG QUALITY

Three-layer evaluation framework

- **KG Quality**

Completeness — % of entities with name, type, and aliases populated.

Consistency — orphan triples and unknown label types flagged. Duplication rate — identical entity names and repeated (subject, predicate, object) triples. Labeling accuracy — entity type validity and coverage across the graph.

- **Structural Health**

Orphan detection — nodes with zero connections that cannot support reasoning chains. Schema compliance — edges checked against ontology-defined allowed relationships. Entity dedup — near-duplicate entities flagged via embedding cosine similarity. Multi-hop connectivity — can the graph support $A \rightarrow B \rightarrow C$ chain reasoning? Aggregated into a 0–100 health score with an interpretable verdict.

- **SFT Pair Quality**

Faithfulness — does the response rely only on graph facts (no hallucination)? Answer relevancy — does the response actually address the instruction? Factual correctness — does the response match the original source triples? Semantic diversity — are the generated instructions varied enough to produce robust training data?

EVALUATION — ABLATION RESULTS

Model B (KG-trained) wins decisively across every metric

Model A = base model (previously fine-tuned, no further training). **Model B** = fine-tuned on SFT pairs generated from our knowledge graph. **Model C** = fine-tuned on QA pairs from the same source documents, but without KG structuring.

Metric	Model A (Base)	Model B (KG)	Model C (Raw)	Winner
Factual Accuracy	2.7%	18.1%	4.2%	B (6.8×)
Multi-hop Accuracy	2.7%	18.1%	4.2%	B
Hallucination Rate	94%	0%	36%	B (zero!)
Consistency Score	0.56	0.83	0.76	B
Avg Response Length	82 words	21 words	24 words	B (concise)

Key insight: Raw-text fine-tuning barely helps (4.2% vs 2.7%). The KG structure is what matters. With **0% hallucination rate**, KG-trained models never fabricate entities — and produce the most concise, focused answers.

⚠ **Caveats:** Results from small_data of 10 vietnamese wikipedia articles. Only 50 test samples with wide confidence intervals. Metrics are heuristic (token-overlap F1, word-count proxies).

Built-in safeguards against hallucination and data risk

As shown earlier: every fact is **provenance-tracked** to its source document, extraction is **ontology-constrained** to reduce hallucination risk, document versions are **preserved in MongoDB** as permanent ground truth, and Neo4j updates are **atomic** — old chunks are deleted and new data merged in a single transaction that rolls back on failure. Here's how each layer reinforces trust:

- **Provenance-backed facts:** Every entity and relationship is traced to its source document via content hash and ingestion run ID — nothing is invented.
- **Ontology-constrained extraction:** Entities and relations are validated against a structured ontology, not free-form generation, reducing hallucination risk.
- **Full audit trail:** MongoDB archives every document version permanently; Neo4j deletions are logged in `archived_chunks` — the graph can be rebuilt from ground truth at any time.
- **Privacy-first design:** Scraper respects robots.txt; MongoDB stores canonical IDs without personal identifying information; no user data is collected.
- **Transparent limitations:** Vietnamese NLP model gaps are documented; domain coverage is tracked per ingestion run; outputs include confidence signals where available.

PRODUCT & USER EXPERIENCE

From raw text to training data — a clean, guided workflow

- **Web dashboard:** Upload sources, configure pipeline parameters, and trigger KG generation from a single interface — no CLI required for day-to-day use.
- **Live graph explorer:** Browse the resulting knowledge graph interactively — search entities, trace relationships, and inspect provenance per node.
- **One-click export:** Download training-ready data in JSON, GraphML, or Neo4j CSV format for direct consumption by fine-tuning and RAG pipelines.
- **Decision-support focus:** ML teams can assess data coverage, spot gaps, and verify factual grounding before committing to expensive training runs.
- **Vietnamese-first interface:** All labels and workflows support Vietnamese language; the system is designed for local AI practitioners, not just English-speaking researchers.

AI-NATIVE INNOVATION

AI isn't a feature — it's the engine

- **ML-powered extraction:** Entity and relation extraction uses transformer models fine-tuned for Vietnamese, with LLM fallback for low-confidence spans — not regex or keyword matching.
- **Semantic deduplication:** Near-duplicate detection uses embedding similarity, not just hash collisions, catching paraphrased content that traditional dedup misses.
- **Vietnamese-first, not translated:** Ontology, extractors, and quality filters are built for Vietnamese from the ground up — not repurposed from English tooling with a language flag.

BUSINESS MODEL & PILOT PATHWAY

From hackathon prototype to real-world adoption

- **Target users:** Vietnamese AI labs building local LLMs, government digitization teams, university research groups, and enterprises needing structured Vietnamese knowledge.
- **Value proposition:** Turn months of manual data curation into hours of automated pipeline execution — produce model-ready structured data at a fraction of the cost.
- **Pilot pathway:** Partner with one Vietnamese university or government agency → run the pipeline on their existing corpus → measure fine-tuning improvement → publish results → expand.

COMPETITIVE ADVANTAGE

Why our approach wins

- **KG beats raw text:** Structured entities and relationships provide higher-density training signals — models learn facts, not just token patterns.
- **KG beats vector DB alone:** Vectors capture similarity but lose explicit relationships; our graph preserves who-connected-to-whom, with evidence provenance.
- **Scalable through graph DB storage:** Neo4j stores the knowledge graph on disk — not in RAM — so the pipeline scales to millions of nodes without exhausting machine memory. A legal web scraper continuously feeds fresh Vietnamese content to keep the graph growing.
- **Dual-store architecture:** MongoDB as permanent ground truth + Neo4j as queryable knowledge graph — no other Vietnamese data pipeline offers this level of auditability.
- **Continuous update loop:** New documents are ingested, old chunks are atomically replaced, and provenance is preserved — the graph stays current without rebuilds.
- **Vietnamese moat:** Purpose-built for Vietnamese language, culture, and data sources — not a generic tool with a language flag. Scrapers, extractors, and ontologies are all Vietnam-native.